aggregate, combine, summarize

data points

Barycenters: Motivation

- In ML, we often need to **aggregate, combine, summarize** data
- We know how to aggregate simple **data points**: e.g., summary statistics (mean, median), clustering, nearest neighbors.
- These are all instances of computing a "center" in Euclidean geometry, e.g. $$x^\star = \arg\min_x \sum_{i=1}^{n} \|x - x_i\|^2 \quad \Rightarrow \quad x^\star = \text{mean}(x_1, \ldots, x_n)\,.$$
- Question: How do we extend this idea of "finding a center" when our objects are probability distributions rather than points?

Barycenters in Probability Space

- Why would we need to do this?